

CSE 112:

Computer Organization
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Quiz III | Set-A

IIIT-Delhi, Winter 2025

Duration : 60 Minutes

Date: Apr 08, 2025

Points: 30

Guidelines

If you found any ambiguity in any of the questions or there appears to be a lack of information, then write an assumption on the answer sheet to explain your side interpretation of the problem and solve accordingly.

FIRST Block Instructions	Description	Operation
addi rd,rs, imm[11:0]	Add immediate	$rd = rs + sext(imm[11:0])$
sub rd, rs1, rs2	Subtract registers	$rd = rs1 - rs2$
add rd, rs1, rs2	Add registers	$rd = rs1 + rs2$
lw rd,#imm[11:0](rs1)	Load Word	$rd = mem(sext(rs1 + imm))$
sw rd,#imm[11:0](rs1)	Store Word	$mem(sext(rs1 + imm)) = rd$
beq rs1, rs2, imm[12:1]	Branch if equal	$PC = PC + sext(imm[12:1], 1'b0)$ if $rs1 == rs2$
beq x0,x0,#0		HALT

Stage Description of a general Five Stage Processor

1. Fetch - Fetches the instructions from instruction memory
2. Decode - Decodes the instructions and type of operations. It also reads the input operands.
3. Execute - Perform the arithmetic and logical operations, determine branches.
4. Memory - Perform the read and write operations at data memory.
5. Writeback - Updates the CPU register value.

[CO3] Problem 1: Control Hazards

[05 Points]

Consider a processor following RISC-V ISA and having a four-stage pipeline with the stages Fetch (F), Decode (D), Execute (E), and **Update (U)**. The processor does not have any data forwarding hardware and branches are determined in the execute stage. You need to determine the CPI (Cycles Per Instruction) of the below-mentioned assembly program.

Update(U) - This stage is a combination of both Memory and Writeback stage. The data memory read and write operations along with writes on CPU registers are performed in this stage.

addi x1, x0, #5
addi x2, x0, #4
addi x3, x0, #1
sub x4, x1, x3
beq x4, x2, END
add x1, x3, x4
sub x4, x2, x3
END: beq x0, x0, #0

[CO3] Problem II: Branch Penalty**[02 Points]**

Consider two processors C, D following RISC-V ISA and having stage description as follows. You need to calculate the Branch Penalty of both the processors.

- (a) Processor C has 37 stages as F1, F2, ... F10, D1, D2, ... D5, E1, E2, ... E9, M1, M2, ... M7, W1, W2, ... W6. The branch operations are determined in the stage E5.
- (b) Processor D has 95 stages as F1, F2, ... F30, D1, D2, ... D20, E1, E2, ... E20, M1, M2, ... M15, W1, W2, ... W10. And, the branch operations are determined in the stage E5.

[CO2] Problem III: Compiler Optimization and Load Delay Slot**[04 Points]**

The compilers are advanced enough to optimize the code to fill the Load Delay slots. Considering a compiler capable of compiling the RISC-V assembly program. You need to optimize and if needed reorganize the below-mentioned assembly code such that the code takes a minimum number of cycles to complete. Assume that the processor has the hardware to support data forwarding.

Assembly Program

```
0x00: Addi x1,x2,x0
0x04: Sub x3,x2,x1
0x08: Lw x4,8(x2)
0x0C: Add x5,x4,x0
0x10: Sub x6,x5,x1
0x14: Add x11,x10,x8
0x18: Add x13,x10,x8
0x1C: Sub x14,x10,x8
```

[CO3] Problem IV: Data Hazards**[03 + 03 Points]**

The timing diagram of a processor following FIRST Block Instructions from RISC-V ISA is shown in figure[1]. You need to execute the below-mentioned assembly program on a five stage microprocessor for the two cases.

- (a) There is no data forwarding hardware in the microprocessor.
- (b) The microprocessor is designed to support data forwarding figure[2].

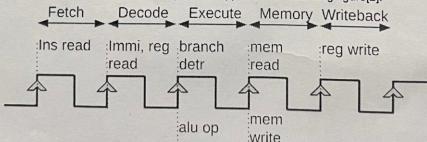


Figure 1: Timing diagram of a Five Stage RISC-V Processor following FIRST BLOCK Instructions.

Assembly Program

0x00: Add x1,x0,#2
 0x04: Lw x3,8(x0)
 0x08: Add x4,x3,x0
 0x0c: Add x5,x4,x0
 0x10: Add x6,x5,x4

[CO3] Problem V: Hazard Control Unit**[09 Points]**

You need to draw the pipeline diagram of the below mentioned assembly program executed on a processor having the micro architecture shown in figure[1]. Thereupon, you need to highlight by creating **CIRCLE** on the stage for data forwarding path **1A** and **SQUARE** on the stages where the data forwarding path **1B** are being utilized for data forwarding.

Assembly Program

Addi x1,x0,#2
 Add x2,x1,x1
 Add x3,x2,x1
 Lw x4,8(x3)
 Add x5,x4,x0
 Add x6,x5,x4
 Lw x7,8(x6)
 Lw x8,8(x7)
 Add x9,x8,x1

[CO3] Problem VI: CPI Analysis**[03+01 Points]**

You need to calculate the speedup in the execution of the mentioned benchmark program when executed on two different processors A and B. Both processors have five stages and operate as per the clock cycle diagram mentioned in figure[2].

Processor_A Miss Predicts 30% of branches and carries no data forwarding hardware capability.

Processor_B Miss Predicts 5% of branches and contains hazard control as shown in figure[2].

Assume 20% of the LOADs in the benchmark program are used by the next/subsequent instruction.

Specification of the Benchmark Program

30% LOAD
 20% STORE
 10% BRANCH
 40% ALU Operation

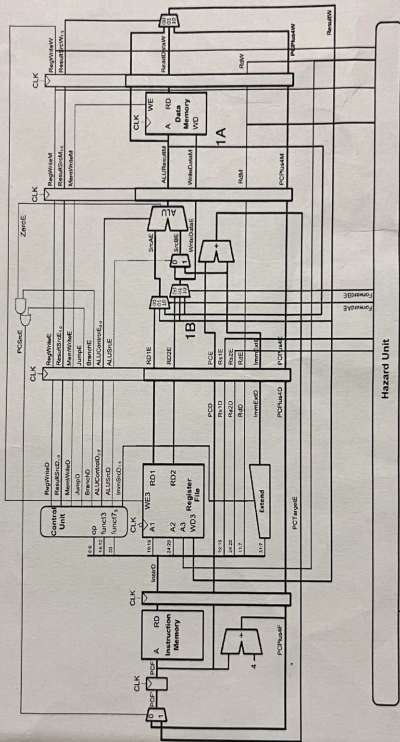


Figure 2: Microarchitecture of Five Stage RISC-V Processor following FIRST BLOCK INSTRUCTIONS.

Image Source: S. L. Harris and D. Harris, "Digital Design and RISC-V Computer Architecture Textbook," 2021 ACM/IEEE Workshop on Computer Architecture Education (WCAE), Raleigh, NC, USA, 2021.